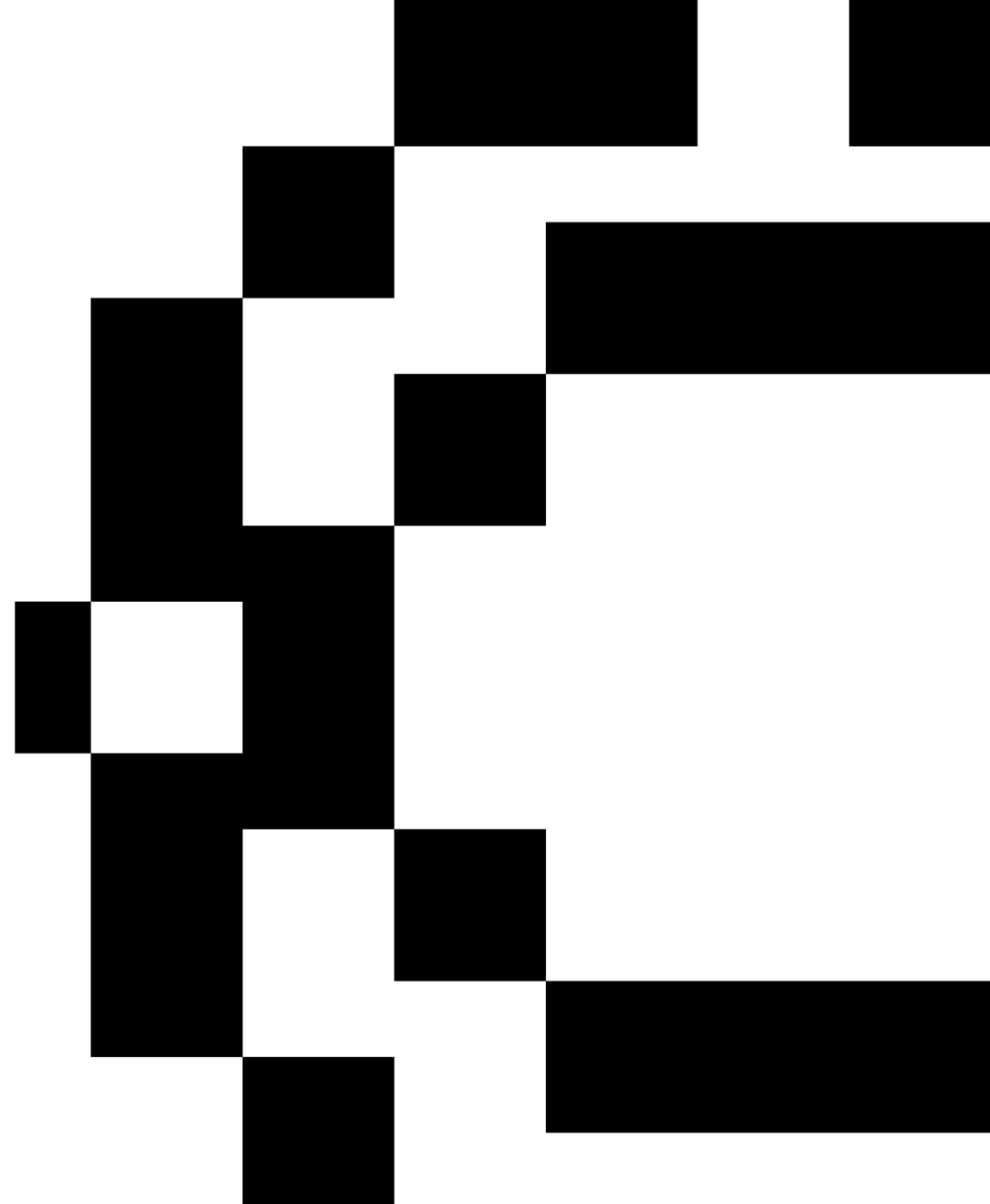


Introduction to Text Mining

Text Mining 2025-2026

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Lecture Plan

01. Introduction to Natural Language Processing (NLP)

02. Challenges of Natural Languages

03. The NLP Pipeline

04. Text Preprocessing Methods

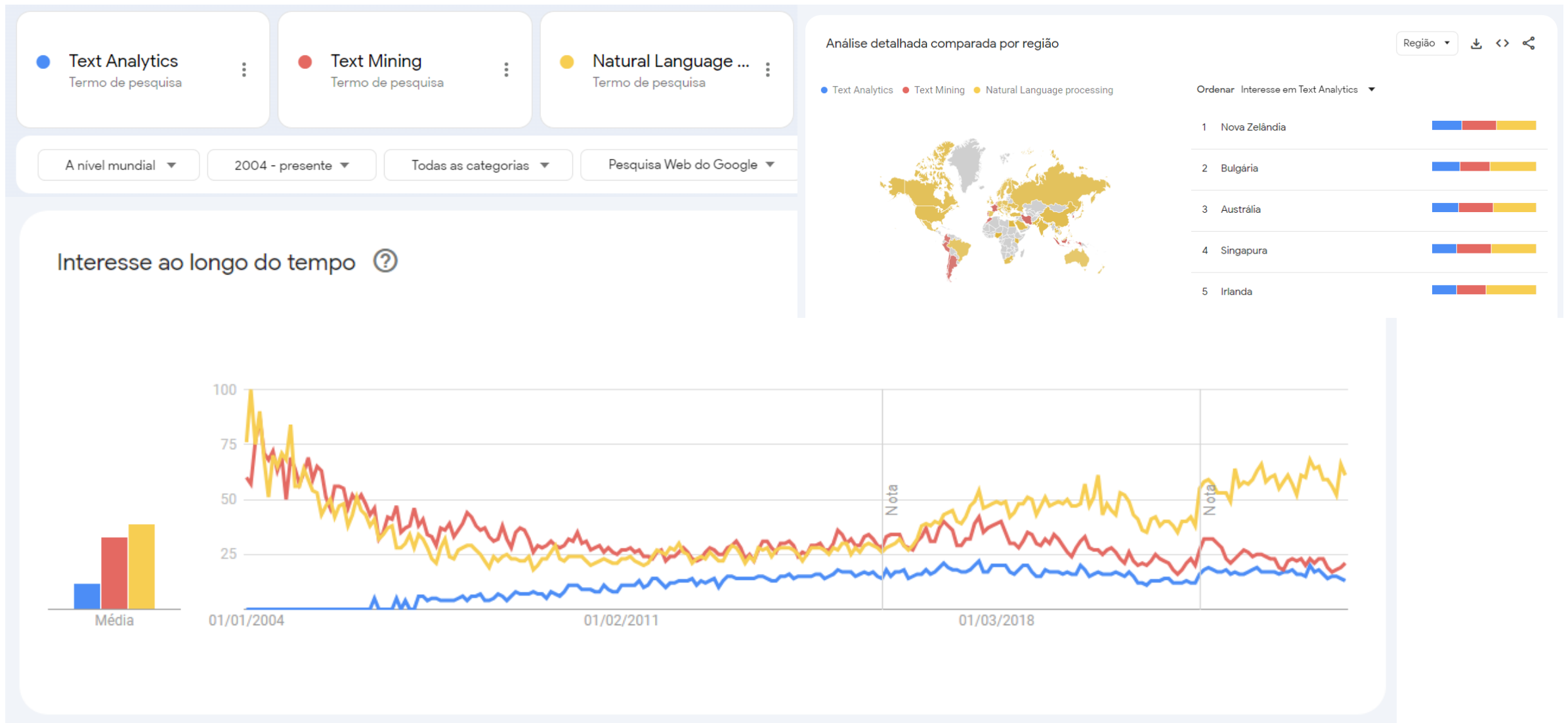


01.

Introduction to NLP

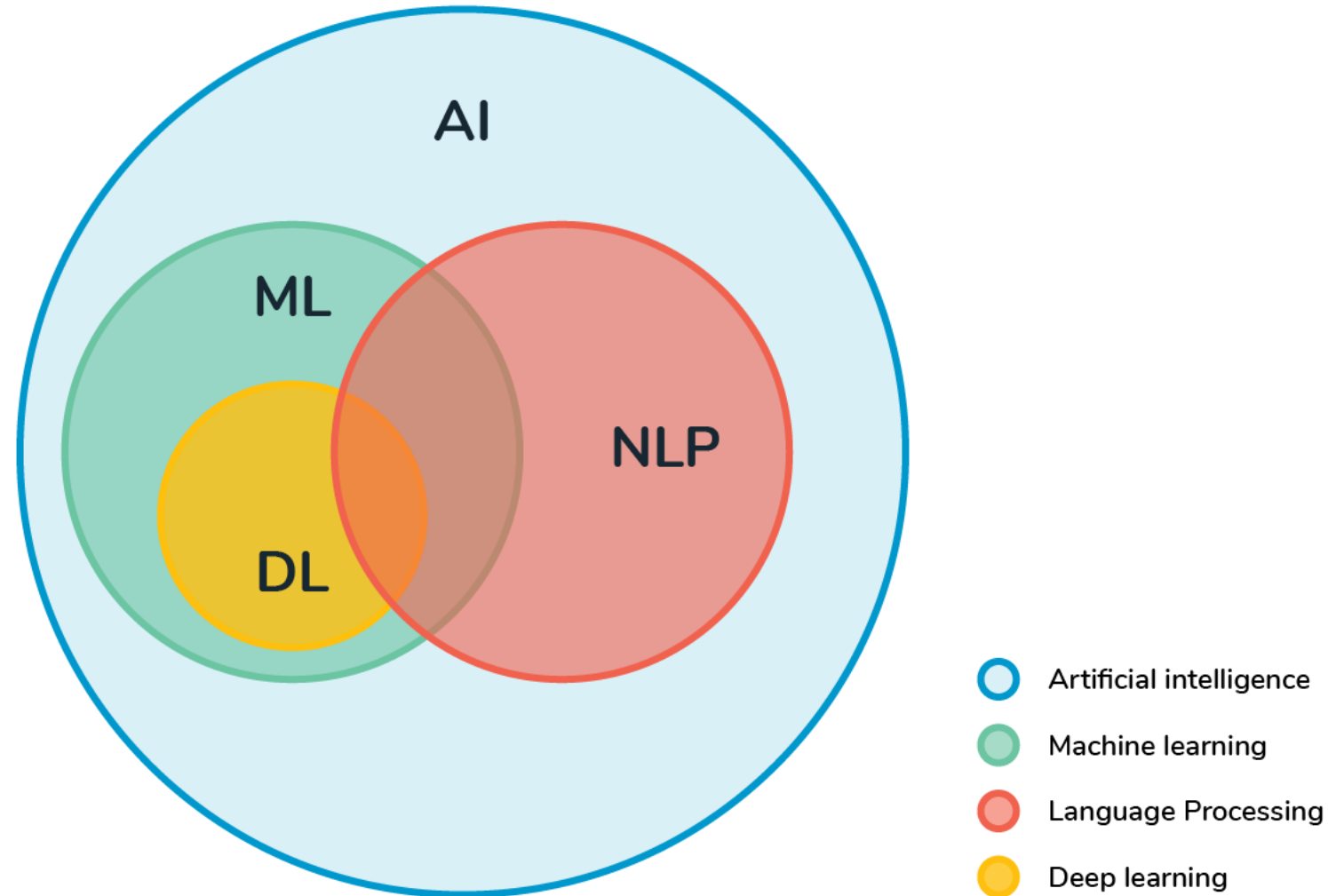
Text Analytics vs Text Mining vs NLP

Introduction to NLP



NLP, AI, Machine Learning & Deep Learning

Introduction to NLP



Definition:

Natural language processing (NLP) is an area of computer science and artificial intelligence that is concerned with the interaction between computers and humans in natural language. The **ultimate goal of NLP is to enable computers to understand language as well as we do (or better 😊).**

1. Machine translation
 2. Dialog systems (chatbots)
 3. Search Engines
 4. Predictive Keyboards and auto correctors
- ...

Everything that deals with text!!!





02. Challenges of Natural Language

What is a Natural Language

Challenges of Natural Language

Natural Language (NL):

1. Grammatical system, with its **own rules**, used by people to communicate
2. Natural **evolution** due to people communication
 - ✓ New words everyday: “bue”, “fixe”, “ups”, “lol”

Examples:

1. English
 2. Portuguese
- ...
- Python??

Definition:

Different **sentences can have the same meaning**; thus, we can say the same thing in different ways. Two sentences with the same meaning are called **paraphrases**.

Examples:

“The president greets the press”

VS

“Trump speaks to the media”

“He has tons of stuff to throw away”

VS

“He needs to get rid of a lot of junk.”

Definition:

A **single sentence can have different meanings**

Dealing with it:

The only way to deal with **ambiguity** is through context!



Often an **NLP system is trained with a corpus** in a specific domain, but it is used in a different domain.

Thus, the system is **confronted with input that he has never seen before**, either because those inputs are **Out-of-Domain (OOD)** or because they contain words **Out-of-Vocabulary (OOV)**.



Pollution



Pollution



Polución



Poluare



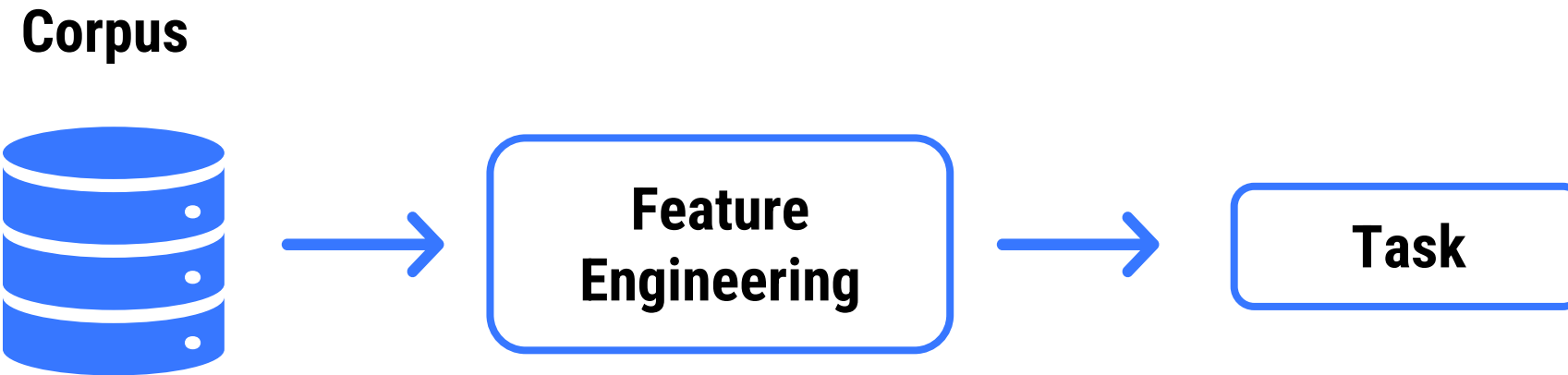
Umweltverschmutzung

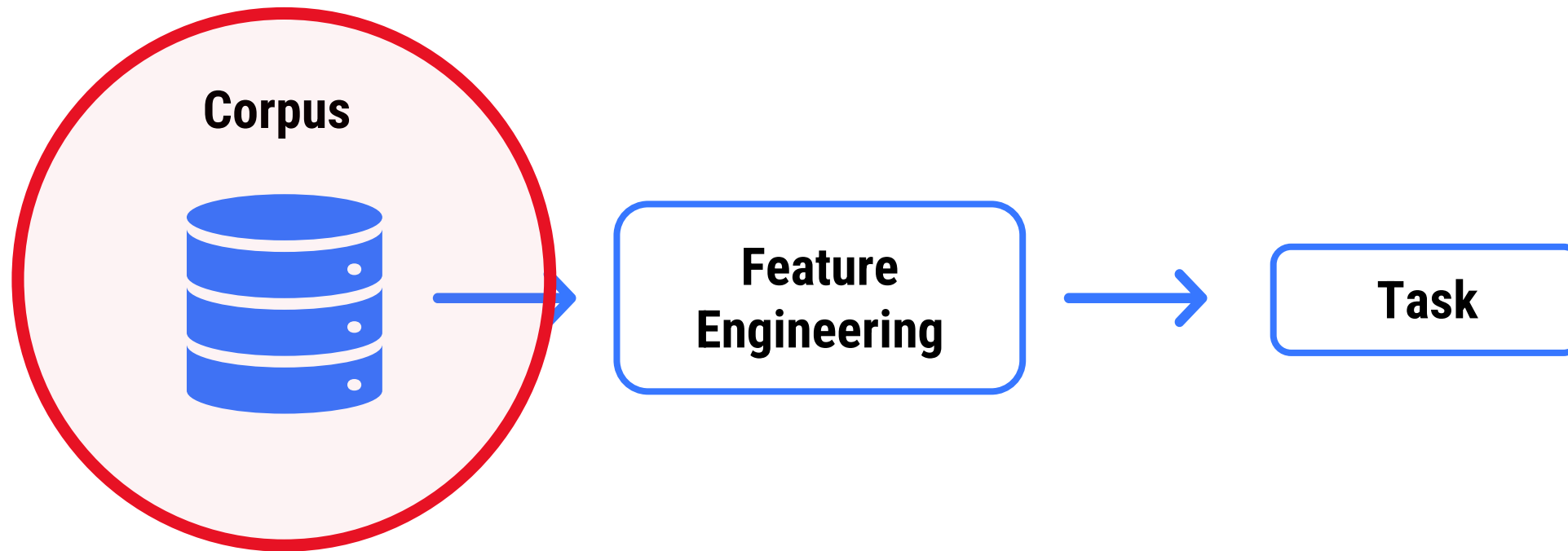




03.

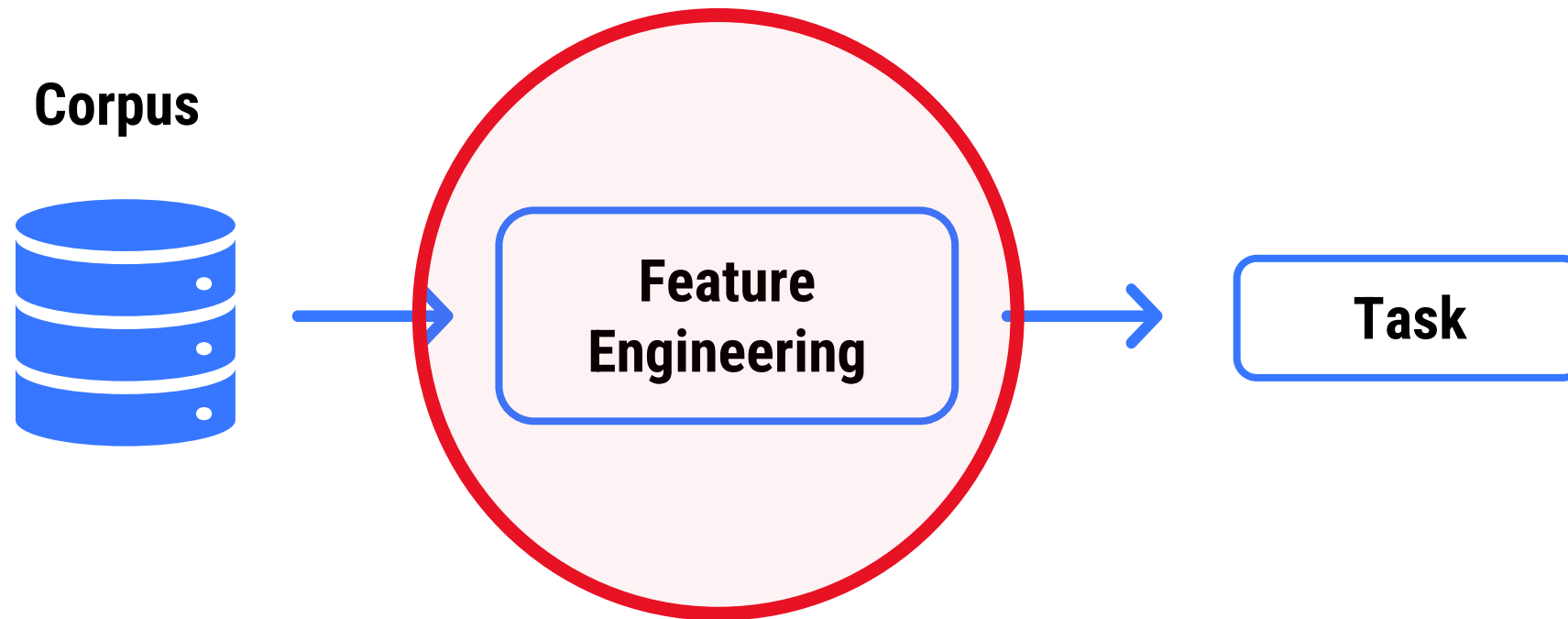
The NLP Pipeline





- A **Corpus** is a collection of **text** organized into datasets.
- A corpus can be made up of everything from **news, recipes, wikipedia pages, academic papers, to novels, movie and tv scripts, and social media posts like tweets.**
- A collection of Corpus is called **Corpora**.

- The first step is to split a corpus into **Train/Validation/Test** or via **K-fold cross validation** (other methods can be considered).
- For **Train/Validation/Test**
 - 80%/10%/10% split for small corpus (<10k samples)
 - More train percentage for bigger corpus.
 - Keep the original always!
 - Your split must be reproducible



How can we represent text?



Bag-of-Words

- Each word is a feature.
- Our **feature space is defined by our vocabulary**
- Documents/pieces of text will be represented as **sparse vectors**.

the dog is on the table

0	0	1	1	0	1	1	1
are	cat	dog	is	now	on	table	the

Bag-of-Words Example

Text 1 – “I love Paris”

Text 2 – “ I live in France”

	France	I	in	live	love	Paris
Text 1						
Text 2						

Bag-of-Words Example

Text 1 – “I love Paris”

Text 2 – “I live in France”

	France	I	in	live	love	Paris
Text 1	0					
Text 2	1					

Bag-of-Words Example

Text 1 – “I love Paris”

Text 2 – “I live in France”

	France	I	in	live	love	Paris
Text 1	0	1				
Text 2	1	1				

Bag-of-Words Example

Text 1 – “I love Paris”

Text 2 – “I live **in** France”

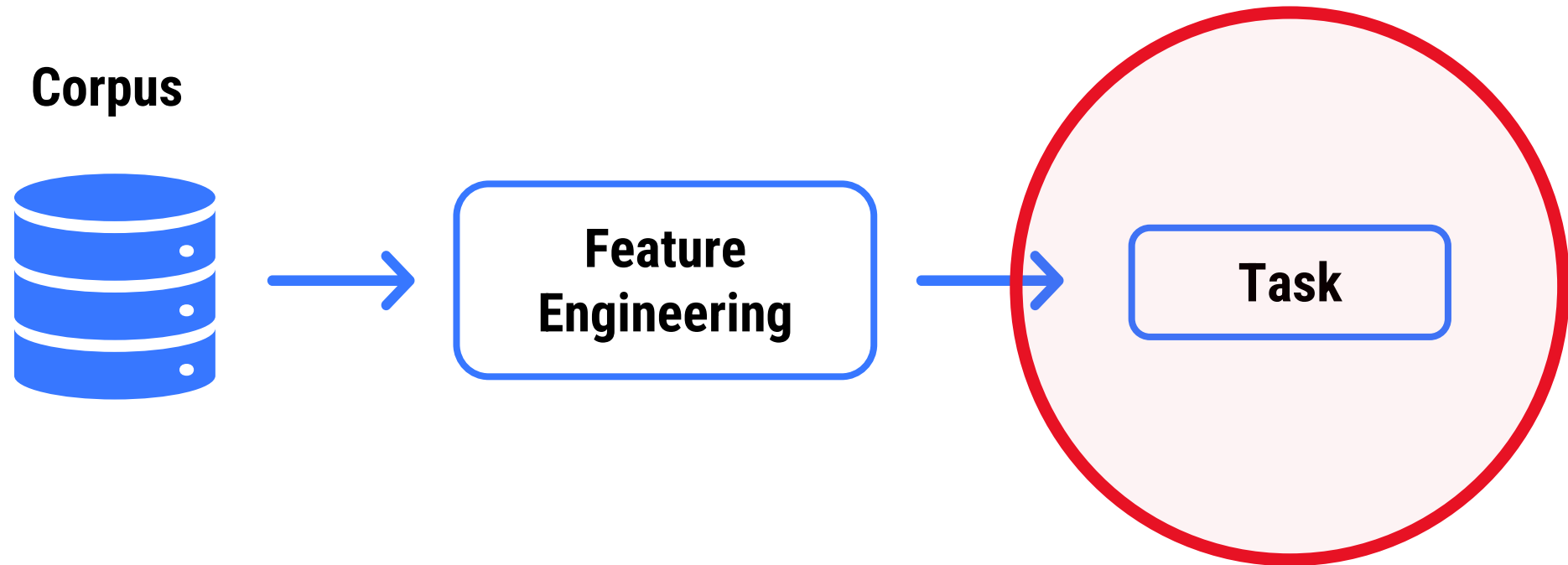
	France	I	in	live	love	Paris
Text 1	0	1	0			
Text 2	1	1	1			

Bag-of-Words Example

Text 1 – “I love Paris”

Text 2 – “ I live in France”

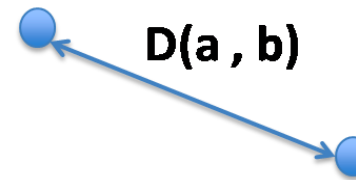
	France	I	in	live	love	Paris
Text 1	0	1	0	0	1	1
Text 2	1	1	1	1	0	0



- Keyword Extraction
- Text Similarity
- Document Classification
- Sentiment Analysis
- Topic Modelling
- Information Retrieval
- Q&A
- Text Generation
- (...)

- Given 2 documents **we can transform them into sparse vectors** and **compare them with simple distance metrics**.
- If they share a lot of words, they will be close to each other.
- This is the **basis for Text Similarity, Document Classification, Sentiment Analysis, Topic Modelling, etc.**

$$D(a, b) = \sqrt{\sum_{i=1}^n (b_i - a_i)^2}$$

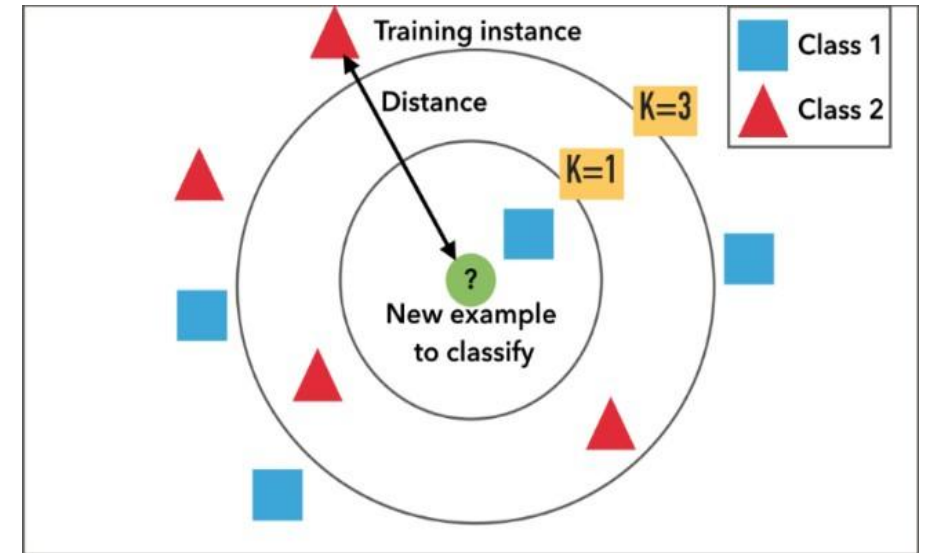


K-Nearest-Neighbour: (our baseline)

The simplest classification algorithm!

We take our new document x , and we represent x in the **same feature space of our training documents**.

Then we **compare x with all training documents** and we **give x the label of the closest** document.





04. Text Preprocessing Methods

Bag-of-Words Example

Text 1 – “I love Paris”

Each word is a feature!

But how to divide sentences into “words”?

Text 2 – “I live in France”

	France	I	in	live	love	Paris
Text 1	0	1	0	0	1	1
Text 2	1	1	1	1	0	0

Tokenization (or word segmentation) is the process of **identifying tokens/features in text**.

This process is **not trivial**:

- United Kingdom ? Is it two words or one?
- Sexta-feira?
- I am vs I'm?

Punctuation can bring several problems to the tokenization.

Typically, we **treat punctuation as individual tokens** but sometimes this can bring problems:

“...” — [. , . , .]

“Dr.” — [Dr, .]

“www.google.com” — [www, . , google, . , com]

vs

Hey! — [Hey, !]

I love NLP. — [I, love, NLP, .]

1. Curse of Dimensionality
2. No Semantic Relationships

1. Curse of Dimensionality

- The **total dimension of the BoW is the vocabulary size**. It can easily **over-fit your model**.
- The remedy is to **use some well-known dimensionality reduction** technique to your input data.

2. No Semantic Relationships

Doesn't consider the semantic relation between words.

- **Word order is discarded**
- **Similar words are treated as different features** and do not share any information...

E.g.: Paris and France should share some information since, for example, they are both locations (see example next)

The Bag-of-Words size equals the size of the vocabulary!

Text 1 – “I love Paris”

Text 2 – “I live in France”

	France	I	in	live	love	Paris	(...)	word n-1	word n
Text 1	0	1	0	0	1	1	...	0	0
Text 2	1	1	1	1	0	0	...	0	0



Some words are not very informative, and **their presence only represents more rules and/or more processing**. Thus, the first thing we can do is to get rid of them.

Examples: on, the, a, of, from, and, that, it, in (...)

“Who **is the** main actor **of** Pulp Fiction”

- [who, main, actor, Pulp Fiction]

Less tokens same meaning!

“What actor played **the** main role **in** Pulp Fiction”

- [what, main, actor, Pulp Fiction, played, role]

Converting everything to **lowercase** also helps **reducing the vocabulary size**. Examples:

“Cats sleep more than cats play.” → [Cats, sleep, more, than, cats, play] → **#words = 6**

“Cats sleep more than cats play.” → [cats, sleep, more, than, play] → **#words = 5**

Sometimes it can go wrong.... Bush (name) != bush (plant)

In some situations, you may want to normalize dates, numbers, names, etc.

Examples:

"03-04-18" vs. "03/04/18"

"John Fitzgerald Kennedy" vs. "John F. Kennedy" vs. "John Kennedy "

Using **regular expressions** we can **convert dates, names and number from one format to another.**

Or, we can **replace them with a default token.** In industry this is an important step for anonymization!

Text 1 – “I love Paris”

Text 2 – “ I live in France”

	France	I	in	live	love	Paris
Text 1	0	1	0	0	1	1
Text 2	1	1	1	1	0	0

$$D(\text{Text 1, Text 2}) = \sqrt{(1-0)^2 + (1-1)^2 + (1-0)^2 + (1-0)^2 + (0-1)^2 + (0-1)^2} = \mathbf{2.24}$$

Text 1 – “I love ~~Paris~~” → “I love **LOCATION**”

Text 2 – “I live in ~~France~~” → “I live in **LOCATION**”

	France	I	in	live	love	Paris
Text 1	0	1	0	0	1	1
Text 2	1	1	1	1	0	0

Text 1 – “I love **LOCATION**”

Text 2 – “ I live in **LOCATION**”

	I	in	live	love	LOCATION
Text 1	1	0	0	1	1
Text 2	1	1	1	0	1

$$D(\text{Text 1}, \text{Text 2}) = \sqrt{(1-1)^2 + (1-0)^2 + (1-0)^2 + (0-1)^2 + (1-1)^2} = 1.73 (<2.24)$$

Transform words in a way that if they represent the same meaning they are captured by the same token. Reduce inflection in language (Inflected Language)

- **Stemming** is the process of removing the last few characters of a given word, to obtain a shorter form, even if that form doesn't have any meaning.
Ex: **Change, changing, changes, changer** becomes **chang**
- **Lemmatization** is the process of turning words into their root word.
Ex: **Change, changing, changes, changer** becomes **change**

Consider only certain type of words (e.g: consider only verbs, nouns, adjectives and adverbs).

The	Dog	Is	Sitting	On	The	Table
×				×	×	
DET	NOUN	VERB	VERB	ADP	DET	NOUN

Corpus



Bag-of-Words

- Tokenization
- Lowercasing
- Stop words
- Regular Expression



- Keyword Extraction
- Text Similarity
- Document Classification
- Sentiment Analysis
- Topic Modelling
- Information Retrieval
- Q&A
- Text Generation



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Thank you!

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